

Document Number: **P0555R0**, ISO/IEC JTC1 SC22 WG21
Audience: LEWG
Date: 2017-01-30
Author: Axel Naumann (axel@cern.ch)

string_view for source_location

Table of Contents

- [Introduction](#)
- [Motivation](#)
- [Proposed wording](#)
- [Acknowledgments](#)
- [Revision history](#)
- [References](#)

I have a great respect for incremental improvement, ...
[followed by words that we don't care about here.]

- Steve Jobs; Interview with The Rolling Stone, 1994

Introduction

Access to the *null-terminated byte string* (NTBS) as returned by some of the member functions of `source_location` is unwieldy in `constexpr` contexts. This paper tries to explain and fix the issue.

Motivation

What `constexpr` strings need is a pointer to the start of the NTBS and its length. It is expected that `constexpr` string classes — whatever their shape will be — can be constructed from those two ingredients. We have a wonderful example for such a `constexpr` string class in [p0259](#). An example for a `constexpr` context for `source_location` and strings is the *Static Reflection* paper ([p0194](#)).

`source_location` is a `constexpr` object: all its state is defined at translation time. All its members are `constexpr`. Yet, building a `constexpr` string object out of the values returned by `file_name()` and `function_name()` is non-trivial, as the NTBS length needs to be determined first. That is traditionally done by `strlen` [`cstring.syn`] which is not marked `constexpr`. Instead, a `constexpr` loop must be written to determine the position of the trailing `\0` character.

To facilitate handling the NTBS returned by `source_location`, this paper proposes to instead return `string_view`, which gives access to the NTBS value for instance through `front()`, and to its length through `length()`, both of which are `constexpr`.

Proposed wording

The wording is relative to [n4529](#). [Insertions](#) and ~~deletions~~ are marked the usual way. Grayish background indicates proposed wording.

In 14.1.1 Header `<experimental/source_location>` synopsis [`reflection.src_loc.synop`] change:

```
// 14.1.3, source_location field access
constexpr uint_least32_t line() const noexcept;
constexpr uint_least32_t column() const noexcept;
constexpr const char*string\_view file_name() const noexcept;
constexpr const char*string\_view function_name() const noexcept;
```

In 14.1.3 **source_location** field access [**reflection.src_loc.fields**] change:

```
constexpr const-char*string_view file_name() const noexcept;
```

Returns: ~~The~~A string_view on the presumed name of the current source file (C++14 §16.8) represented ~~by this object~~ as an NTBS.

```
constexpr const-char*string_view function_name() const noexcept;
```

Returns: If this object represents a position in the body of a function, returns a string_view on an implementation-defined NTBS that should correspond to the function name. Otherwise, returns an empty ~~string~~string_view.